

# Customer Support Module

## Across Industries

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*A beginner-friendly guide covering 12 industries,  
AI tools, channels, KPIs & real-world platforms*





## AGENDA

*What we'll cover  
in 20 minutes*

|           |                    |
|-----------|--------------------|
| 0-2 min   | Introduction       |
| 2-10 min  | 12 Industries      |
| 10-13 min | Common Features    |
| 13-16 min | Trends & Future    |
| 16-20 min | Architecture & Q&A |

# Today's Roadmap

0  
1

## What is a Customer Support Module?

Definition + purpose

0  
2

## 12 Industries Deep Dive

Use cases, channels, KPIs, tools for each

0  
3

## Features All Industries Share

Ticketing, AI bots, SLA, analytics

0  
4

## How Industries Differ

Compliance, channels, complexity

0  
5

## Comparison Table

Side-by-side industry snapshot

0  
6

## Current Trends

AI, omnichannel, proactive support

0  
7

## Future of Customer Support

Autonomous AI, AR, blockchain

0  
8

## Reference Architecture

How all pieces fit together

# What is a Customer Support Module?



Think of it as the software "brain" behind all customer help — it receives questions, routes them to the right team, tracks progress, and measures success.



## Receive

Collects all customer requests from any channel (chat, email, phone)



## Route

Sends each ticket to the right person or automated bot



## Resolve

Agent or AI answers the issue and closes the case



## Measure

Tracks CSAT, speed, and quality to improve over time



*This module exists in EVERY industry — the channels and rules change, but the core logic stays the same!*

# 12 Industries We'll Cover

*Each industry has its own unique challenges, tools, and support style*



## E-Commerce

High-volume, fast returns



## Banking

Secure & compliant



## Healthcare

HIPAA, empathy-first



## Telecom

Millions of calls daily



## Travel

Real-time disruptions



## Insurance

Claims & fraud detection



## Education

Students & parents



## Government

Citizen services



## Logistics

Live tracking queries



## SaaS

Tech-savvy users



## Automotive

Dealers & roadside



## Energy

Outage management

# Industries 1 & 2 — E-Commerce & Banking



## E-Commerce & Retail

- **Top Issues:** Lost orders, refund delays, checkout errors
- **Channels:** Live chat, bot, email, social media
- **AI Feature:** Chatbot deflects 70%+ of queries
- **Key Tools:** Zendesk, Gorgias, Intercom
- **Top KPI:** CSAT >85%, FCR >75%
- **Unique:** Black Friday volume spikes ▲



## Banking & Finance

- **Top Issues:** Fraud, failed transfers, loan denials
- **Channels:** Secure IVR, in-app chat, branch walk-in
- **AI Feature:** Fraud detection + voice biometrics
- **Key Tools:** Salesforce FSC, Genesys, Avaya
- **Top KPI:** FCR >80%, ASA <30 sec
- **Unique:** 100% regulatory compliance required ⚖️

# Industries 3 & 4 — Healthcare & Telecom



## Healthcare

- ▶ **Top Issues:** Billing errors, long waits, HIPAA concerns
- ▶ **Channels:** Phone (60%+), patient portal, telemedicine
- ▶ **AI Feature:** Symptom checker bots, no-show prediction
- ▶ **Key Tools:** Epic MyChart, Salesforce Health Cloud
- ▶ **Top KPI:** Patient Satisfaction (HCAHPS), <2 min wait
- ▶ **Unique:** Emotionally charged — need empathy first 🧡



## Telecom

- ▶ **Top Issues:** Bill shock, network outages, plan confusion
- ▶ **Channels:** IVR (dominant), mobile app, retail store
- ▶ **AI Feature:** Smart IVR routing, churn prediction AI
- ▶ **Key Tools:** Genesys, Amdocs, NICE CXone
- ▶ **Top KPI:** FCR >75%, Repeat Contact Rate <15%
- ▶ **Unique:** Millions of contacts — must automate heavily 🤖

# Industries 5 & 6 — Travel & Insurance



## Travel & Hospitality

- **Top Issues:** Cancellations, lost baggage, overbooking
- **Channels:** Phone (for crises!), mobile app, WhatsApp
- **AI Feature:** Delay prediction + proactive rebooking bot
- **Key Tools:** Zendesk, Oracle Hospitality, ReviewPro
- **Top KPI:** NPS >45, Refund in <5 business days
- **Unique:** 24/7 global ops — across time zones 🌍



## Insurance

- **Top Issues:** Claim denial, slow payouts, policy confusion
- **Channels:** Phone, email, web portal, video claims
- **AI Feature:** Computer vision for damage assessment
- **Key Tools:** Guidewire ClaimCenter, Salesforce Insurance
- **Top KPI:** Claims resolved in <15 days, FNOL accuracy
- **Unique:** Customers are stressed — handle with care 💚

# Industries 7 & 8 — Education & Government



## Education & EdTech

- ▶ **Top Issues:** Platform downtime, fee delays, grade disputes
- ▶ **Channels:** Email, student portal, WhatsApp, phone
- ▶ **AI Feature:** Dropout prediction + chatbot for admissions
- ▶ **Key Tools:** Freshdesk, Salesforce Edu Cloud, Intercom
- ▶ **Top KPI:** Query resolved <24 hrs, Student Satisfaction
- ▶ **Unique:** Spike during exam & admission season 



## Government Services

- ▶ **Top Issues:** Slow processing, unclear procedures, red tape
- ▶ **Channels:** Phone (toll-free), web portal, physical centers
- ▶ **AI Feature:** Grievance auto-categorization via NLP
- ▶ **Key Tools:** ServiceNow, CPGRAMS (India), Salesforce Gov Cloud
- ▶ **Top KPI:** Grievance Disposal >90%, CSI score
- ▶ **Unique:** Must serve ALL citizens — no digital exclusion 🙄



# Industries 9 & 10 — Logistics & SaaS



## Logistics & Supply Chain

- **Top Issues:** Lost packages, customs delays, last-mile fails
- **Channels:** Mobile app tracking, SMS, WhatsApp, email
- **AI Feature:** ML-based delay prediction + damage detection
- **Key Tools:** AfterShip, Zendesk, Oracle Field Service
- **Top KPI:** On-Time Delivery >95%, FCR for delivery issues
- **Unique:** Real-time coordination with 50+ carrier partners 🌐



## SaaS (Software as a Service)

- **Top Issues:** Onboarding confusion, API errors, billing disputes
- **Channels:** In-app chat (primary!), community forum, email
- **AI Feature:** Churn prediction via usage signals + AI doc bot
- **Key Tools:** Intercom, Zendesk, PagerDuty, Statuspage
- **Top KPI:** MTTR, CSAT per plan tier, ticket deflection rate
- **Unique:** Must support free users cheaply AND enterprises 📦

# Industries 11 & 12 — Automotive & Energy



## Automotive

- **Top Issues:** Service delays, warranty rejections, EV tech issues
- **Channels:** Phone (roadside!), dealer CRM, mobile app
- **AI Feature:** Predictive maintenance from telematics/IoT data
- **Key Tools:** Salesforce Automotive Cloud, CDK Global
- **Top KPI:** Roadside Response <30 min, Warranty Claim Time
- **Unique:** Support spans 1000s of independent dealers 



## Energy & Utilities

- **Top Issues:** Outages, wrong bills, smart meter malfunctions
- **Channels:** Phone + IVR (primary), SMS alerts, mobile app
- **AI Feature:** Outage prediction from weather + grid data
- **Key Tools:** Oracle CC&B, SAP IS-U, Salesforce Utilities Cloud
- **Top KPI:** SAIDI (outage duration), Bill Dispute Resolution
- **Unique:** Floods support center during storm outages 

# Industry Comparison — At a Glance

| Industry                                                                                   | Main Channel | AI Level                                                                                   | Top Tool        | Challenge          |
|--------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------------|-----------------|--------------------|
|  E-Commerce | Chat / Bot   |  High     | Zendesk/Gorgias | Volume spikes      |
|  Banking    | Phone / IVR  |  High     | Salesforce FSC  | Compliance 100%    |
|  Healthcare | Phone/Portal |  Medium   | Epic MyChart    | HIPAA rules        |
|  Telecom    | IVR / App    |  High     | Genesys/Amdocs  | Agent attrition    |
|  Travel     | Phone / App  |  Medium   | Zendesk         | 24/7 global ops    |
|  Insurance  | Phone/Email  |  Medium   | Guidewire       | Fraud + compliance |
|  Education  | Email / Chat |  Medium   | Freshdesk       | Seasonal spikes    |
|  Government | Phone/Portal |  Low      | ServiceNow      | Legacy systems     |
|  Logistics  | App / SMS    |  High     | AfterShip       | 50+ carriers       |
|  SaaS       | In-App Chat  |  V.High   | Intercom        | Free tier cost     |
|  Automotive | Phone / App  |  Medium   | SF Auto Cloud   | Dealer network     |
|  Energy   | Phone / IVR  |  Medium | Oracle CC&B     | Mass outage mgmt   |

# Features ALL Industries Share

*Regardless of the industry, every support system has these 8 building blocks:*



## Omnichannel Ticketing

One inbox for all channels



## Customer 360° View

Full profile in one screen



## Workflow Engine

Auto-route & escalate tickets



## Knowledge Base

FAQ for customers & agents



## AI Chatbot

Handles 60-80% of Tier-1 queries



## Analytics Dashboard

Real-time KPI monitoring



## SLA Management

Tracks response & resolve time



## CSAT Feedback

Post-interaction surveys

# How Industries Differ — 6 Key Dimensions



## Regulations

Low → E-Commerce, SaaS

High → Banking, Healthcare, Insurance, Government

💡 More regulations = more secure channels + audit trails



## AI Maturity

Low → Government (legacy)

High → SaaS, E-Commerce

💡 SaaS leads in AI adoption; Govt lags most



## Integrations

Low → EdTech (basic LMS)

High → Healthcare (EHR), Telecom (BSS/OSS)

💡 Complex integrations = longer build time



## Main Channel

Low → SaaS = In-App Chat

High → Banking/Telecom = IVR/Phone

💡 Channel choice shapes the entire tech stack



## Emotional Stake

Low → E-Commerce, SaaS

High → Healthcare, Insurance, Travel

💡 Higher stakes = more empathy training needed



## B2C vs B2B

Low → Pure B2C: Utilities, Telecom

High → Significant B2B: SaaS, Logistics

💡 B2B needs dedicated accounts + custom SLAs

# Current Trends Shaping Customer Support

*These are the biggest shifts happening RIGHT NOW in the industry (2024–2025)*



## Generative AI (LLMs)

### GPT/Claude in support

AI writes responses, summarizes tickets, suggests articles. Reduces AHT by 30–50%



## Omnichannel

### No more silos!

Same context flows from WhatsApp → Chat → Phone. No customer repeats themselves



## Proactive Support

### Detect before complaint

AI spots a delivery delay and messages the customer BEFORE they call in



## Self-Service Boom

### Bots handle 60–80%

Customers prefer to solve issues themselves. Investment in smart knowledge bases growing fast



## Sentiment Analysis

### Emotion AI in real-time

Supervisor gets alert when a customer conversation turns angry. Instant intervention



## AI + Human Hybrid

### AI handles Tier 1

Industry agrees: AI for volume, humans for complex/emotional. Not replacement, augmentation

# Future of Customer Support (Next 3–5 Years)



## Autonomous AI Agents



AI handles an entire support case end-to-end — no human needed. Issues a refund, reschedules a booking, updates an account. All automatically.



## Augmented Reality (AR)

### Support



You point your phone camera at a broken device. The support agent sees your screen and guides you with live visual overlays. No more 'turn it off and on again'.



## Blockchain Audit Trails



Insurance claims, bank disputes — all recorded on an immutable blockchain. Automatic payouts when conditions are met (smart contracts). Zero dispute delays.



## Explainable AI (XAI)



Regulators now DEMAND that AI explains its decisions. 'Your claim was denied because...' — AI must show its reasoning, not just output a result.



## Total Experience (TX)



Merging Customer Experience + Employee Experience + User Experience into ONE unified design. Happy agents = happy customers. Both journeys redesigned together.

# Reference Architecture — How It All Works Together

## LAYER 1 — Customer Channels

 Mobile |  Chat |  Phone |  Email |  Social |  In-Person



## LAYER 2 — Omnichannel Gateway + Bot Interception

All contacts enter here → AI Bot tries to resolve → Unresolved → Route to agent



## LAYER 3 — Core Platform (Ticketing • Workflow • Knowledge Base • Agent Desktop)

Ticket created → Priority set → Agent sees 360° customer view → Resolves & closes



## LAYER 4 — AI & Intelligence (NLP • Sentiment • Predictions • Voice Analytics)

Works horizontally across ALL layers — flags anger, routes smart, predicts churn



## LAYER 5 — Integrations (CRM • ERP • Payment • Industry-Specific Systems)

Pulls order data, account info, medical records, vehicle VIN — into one view for agent



## LAYER 6 — Reporting & Governance (KPIs • SLA • Audit • Privacy)

Management sees real-time dashboards. Regulators see audit logs. All data collected here.



# Key Metrics — What Do All These Terms Mean?

**CSAT**

## Customer Satisfaction Score

After the support call ends, did the customer rate it 4 or 5 out of 5? Target: >85%

**FCR**

## First Contact Resolution

Did we solve the problem in just ONE contact? No callbacks needed. Target: >75%

**AHT**

## Average Handle Time

How long does one support interaction take? Shorter = more efficient. Target: <5-7 min

**NPS**

## Net Promoter Score

Would you recommend us to a friend? Score 0-10. Above 7 = promoter. Target: >45

**SLA**

## Service Level Agreement

A promise: we WILL respond in X minutes and resolve in Y hours. If we miss it = breach

**MTTR**

## Mean Time to Resolution

On average, how long does it take to fully fix an issue? Lower is better

**CES**

## Customer Effort Score

How EASY was it to solve your problem? Easier = better loyalty. Scale of 1-7

**ASA**

## Average Speed to Answer

How many seconds before a human picks up the phone? Banking target: <30 seconds

# Real-World Tools Used Across Industries

## Ticketing & CRM

- Zendesk — Universal, every industry uses it
- Freshdesk — Great for SMBs & growing teams
- Salesforce Service Cloud — Enterprise standard
- Intercom — SaaS & product-led companies

## Contact Center

- Genesys Cloud CX — Large enterprise voice
- NICE CXone — AI-powered call centers
- Avaya — Telecom & banking IVR systems
- Twilio — Build-your-own voice & SMS

## Industry-Specific

- Guidewire — Insurance claims management
- Epic MyChart — Healthcare patient portal
- Amdocs — Telecom BSS/OSS support
- Oracle CC&B — Utilities billing & support

## AI & Automation

- Nuance — Voice AI & IVR intelligence
- Liveperson — Conversational AI platform
- Sprinklr — Social media + AI insights
- AfterShip — Logistics tracking & notifications

# Key Takeaways

*What to remember from this presentation*

01

## **Same core, different skin**

All 12 industries use the same 8 features — the rules, channels, and regulations differ

02

## **AI is not replacing agents**

AI handles 60–80% of Tier-1 queries; humans handle the hard, emotional, complex stuff

03

## **Channel = industry personality**

Banking = IVR. SaaS = In-app chat. Telecom = Mobile app. Channel tells you everything

04

## **KPIs tell the health story**

CSAT, FCR, AHT, NPS — these numbers reveal if your support is working or broken

05

## **Omnichannel is the standard now**

Customers expect seamless experience across all channels. Silos are a competitive disadvantage

06

## **Future = Proactive, not Reactive**

The best support teams won't wait for complaints — they'll fix problems before you even notice